Emulator evaluation results

84 records, read from this directory. Bars show the median of the runs of a configuration; dots show each run. Higher is better unless a chart says otherwise.

84	3	4	2026-11-13
result records	hosts: Linux x86-64, Windows x86-64, macOS ARM64	emulator modes	latest record (UTC)

Hosts: Linux x86-64 | Windows x86-64 | macOS ARM64 — Values: Measured

Hosts

The machines the results were measured on, as the records describe them.

Linux x86-64

Machine	The same PC as windows-x86_64, dual boot
CPU	Intel(R) Core(TM) i7-10700K CPU @ 3.80GHz
Logical CPUs	16
Memory	31 GiB
GPU	NVIDIA GeForce RTX 4070 (driver nvidia 595.99.02)
Operating system	Ubuntu 24.04.5 LTS (kernel 6.8.0-146-generic); Ubuntu 24.04.5 LTS (kernel 6.8.0-149-generic)
VM runtime	QEMU 8.2.2 (Debian 1:8.2.2+ds-0ubuntu1.18)
Container runtime	docker client 20.10.18, engine 20.10.18
Container engine	16 CPUs, 31 GiB (Ubuntu 24.04.5 LTS)
Records	46, 2026-11-12

Windows x86-64

Machine	The same PC as linux-x86_64, dual boot
CPU	Intel(R) Core(TM) i7-10700K CPU @ 3.80GHz
Logical CPUs	16
Memory	32 GiB
GPU	NVIDIA GeForce RTX 4070 – stated, not in the records
Operating system	10.0.26300
VM runtime	QEMU 11.1.0 (v11.1.0-12130-ge470268ff4)
Container runtime	podman client 5.8.9, engine 5.8.10
Container engine	16 CPUs, 16 GiB (fedora)
Records	2, 2026-11-11

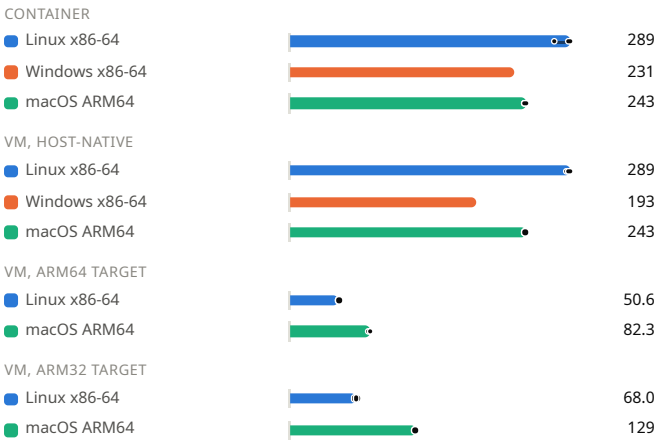
macOS ARM64

Machine	Mac mini, Apple M6, 24 GB
CPU	Apple M6
Logical CPUs	12
Memory	24 GiB
GPU	Apple M6
Operating system	27.2; 27.2.1
VM runtime	QEMU 11.1.3
Container runtime	docker client 29.9.3, engine 29.5.2
Container engine	6 CPUs, 7.7 GiB (Ubuntu 24.04.4 LTS)
Records	36, 2026-11-11 to 2026-11-13

CPU

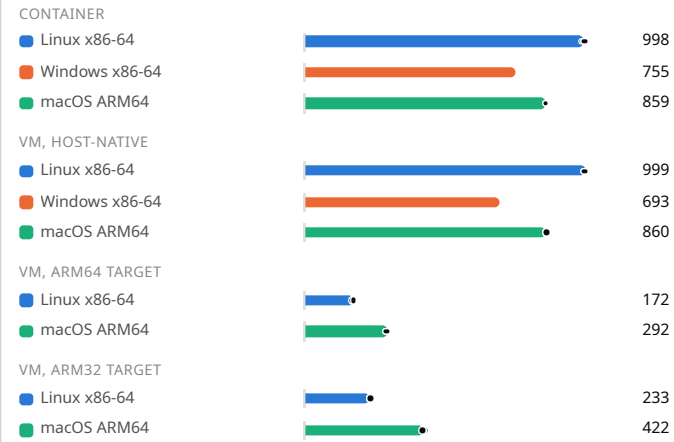
SHA-256, one thread

MiB/s



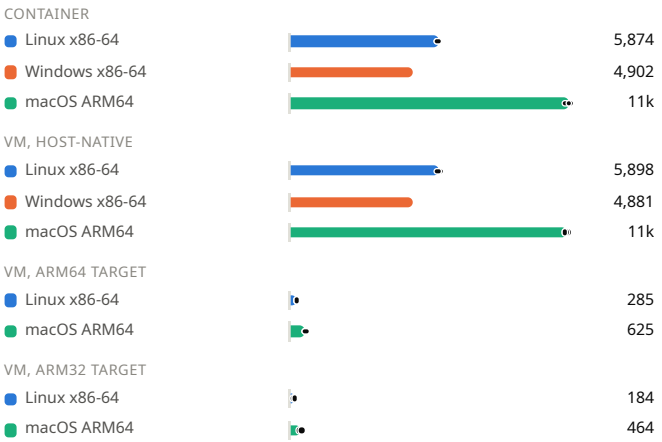
SHA-256, all threads

MiB/s



Matrix multiply (double)

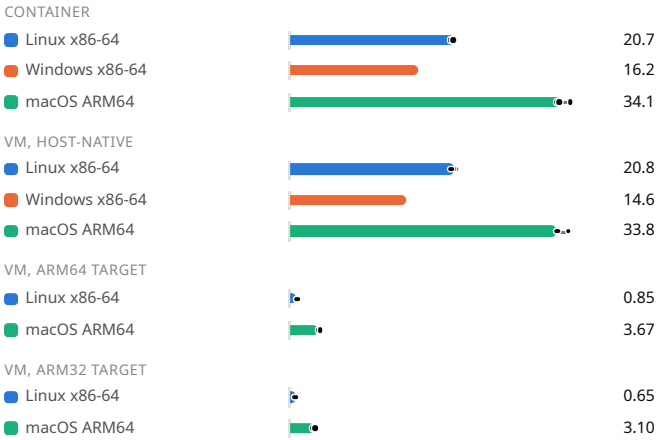
Mflop/s



Memory allocation

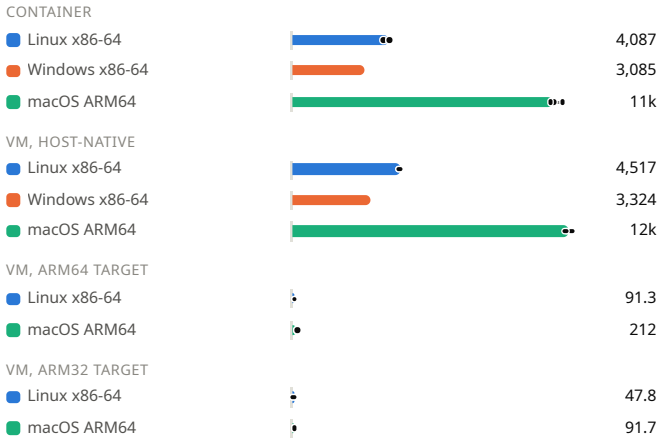
malloc/free churn

Mops/s



First touch of new pages

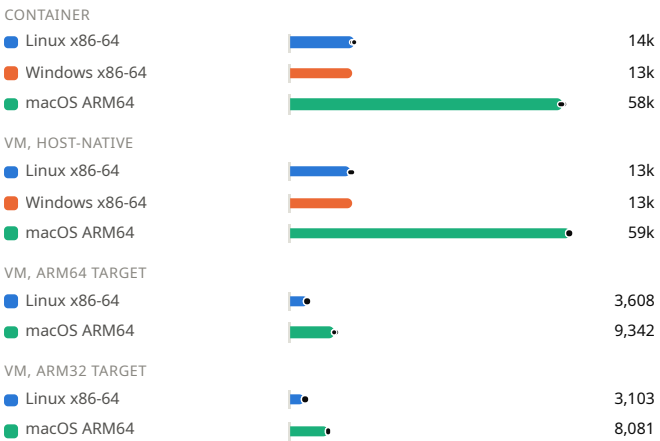
MiB/s



Memory bandwidth

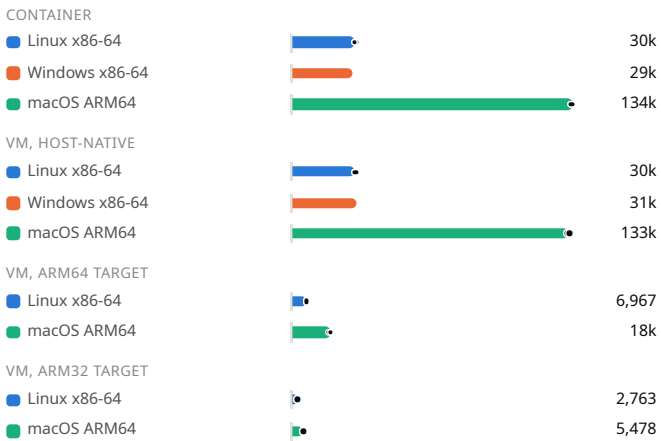
memcpy

MiB/s



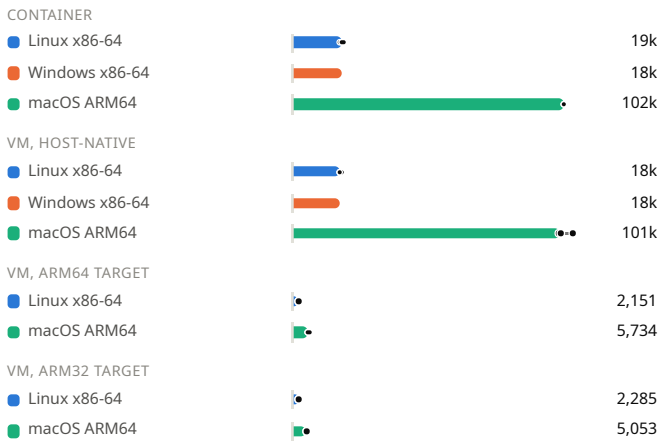
memset

MiB/s



STREAM triad

MiB/s

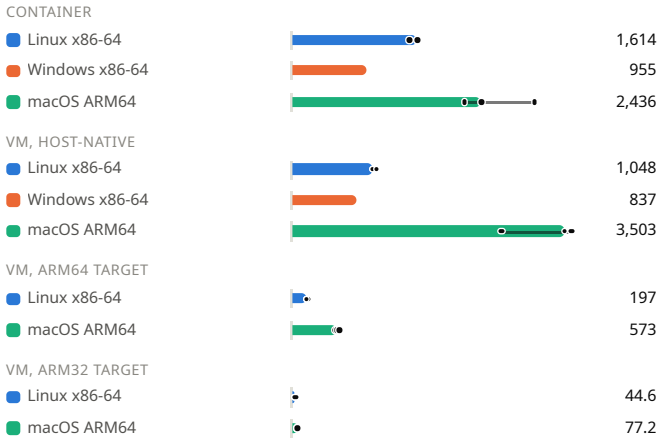


I/O

Sequential read
MiB/s



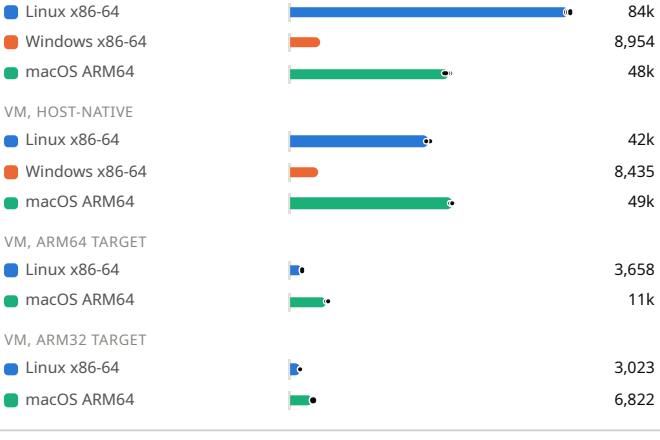
Sequential write
MiB/s



Random read, 4 KiB
IOPS



Random write, 4 KiB
IOPS



Synchronous write, 4 KiB
IOPS

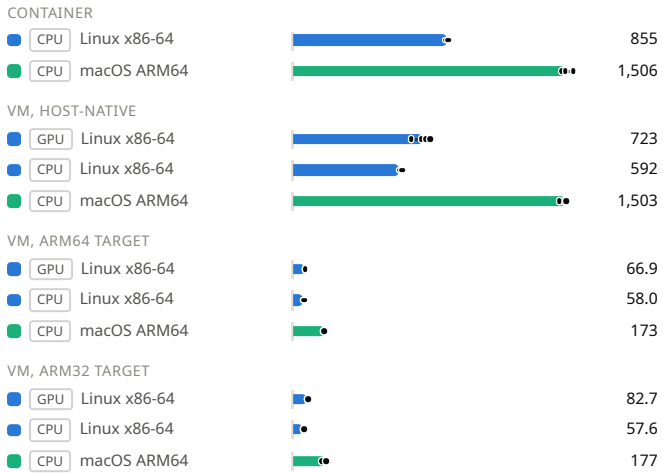


Graphics

“GPU” rows had graphics mapped to the host's GPU; “CPU” rows had them emulated: rendered by the target's CPU. Relative values compare with the emulated container on the same host.

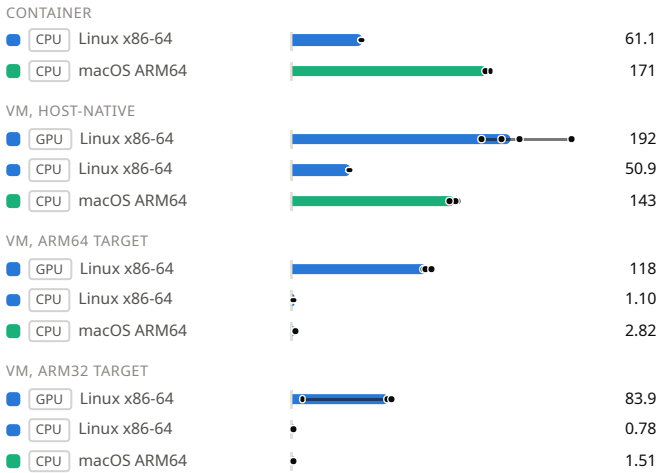
Wayland (CPU-drawn buffers) · 1920x1080

frames/s



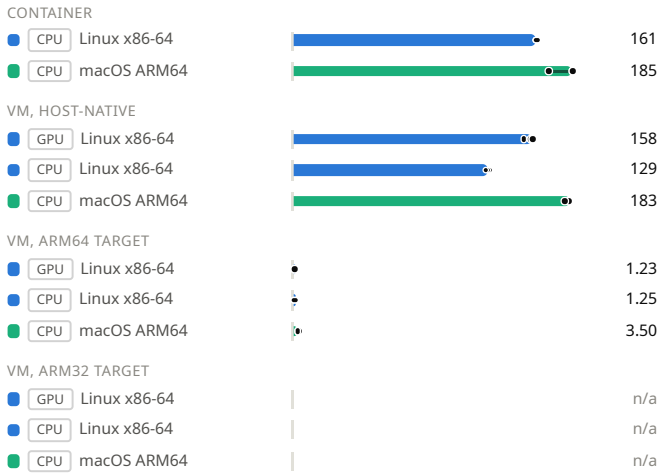
OpenGL ES · 1920x1080

frames/s



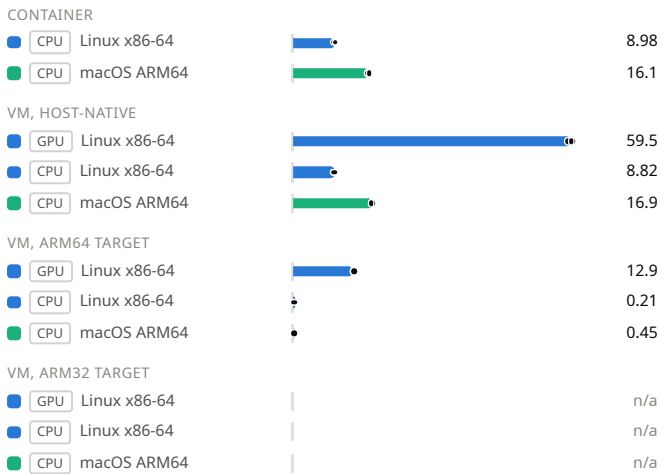
Vulkan · 1920x1080

frames/s



Flutter · 1920x1080

frames/s



Audio and video decoding

Decode speed is media time decoded per second (higher is better). Dropped frames are counted over a 125-frame stream played in real time (lower is better); GPU and CPU say how the display's graphics were provided.

Video decode: AVC in MSE fragments · 1280x720

x realtime



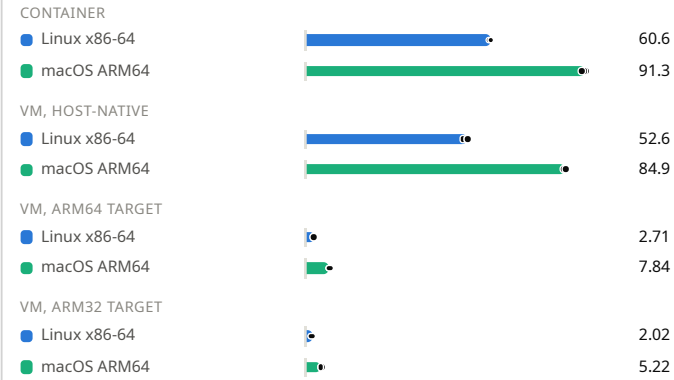
Video decode: AVC in MP2TS · 1280x720

x realtime



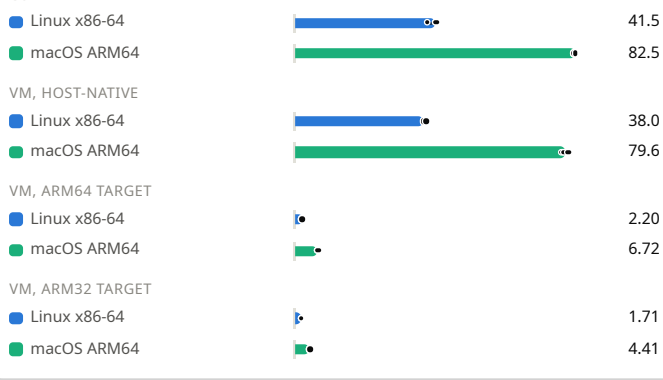
Video decode: AVC in MP4 · 1280x720

x realtime



Video decode: HEVC in MSE fragments · 1280x720

x realtime



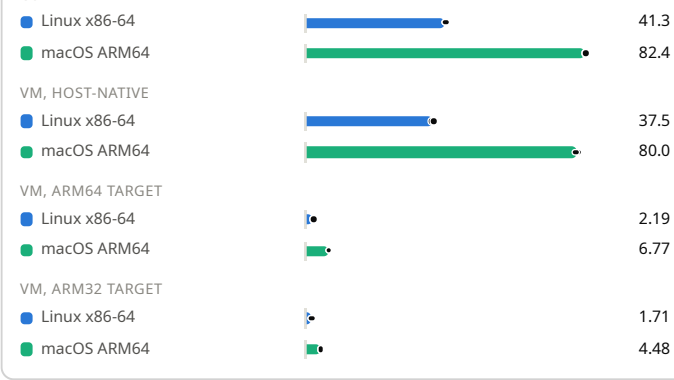
Video decode: HEVC in MP2TS · 1280x720

x realtime



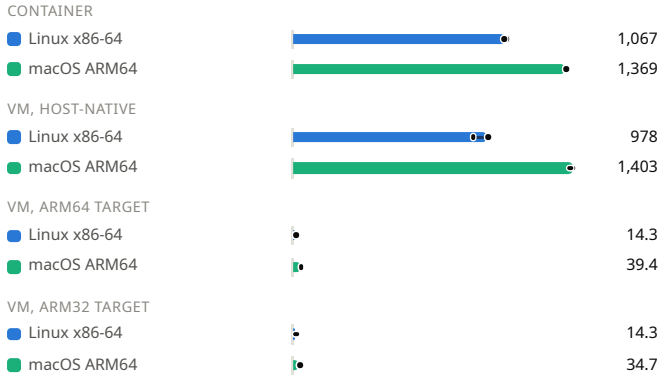
Video decode: HEVC in MP4 · 1280x720

x realtime




Video decode: MPEG-2 in MSE fragments · 1280x720
x realtime
CONTAINER
Linux x86-64 134
macOS ARM64 189
VM, HOST-NATIVE
Linux x86-64 114
macOS ARM64 209
VM, ARM64 TARGET
Linux x86-64 7.06
macOS ARM64 21.4
VM, ARM32 TARGET
Linux x86-64 5.22
macOS ARM64 13.8
Video decode: MPEG-2 in MP2TS · 1280x720
x realtime
CONTAINER
Linux x86-64 132
macOS ARM64 195
VM, HOST-NATIVE
Linux x86-64 119
macOS ARM64 211
VM, ARM64 TARGET
Linux x86-64 7.10
macOS ARM64 20.8
VM, ARM32 TARGET
Linux x86-64 5.34
macOS ARM64 14.0
Video decode: MPEG-2 in MP4 · 1280x720
x realtime
CONTAINER
Linux x86-64 134
macOS ARM64 192
VM, HOST-NATIVE
Linux x86-64 116
macOS ARM64 205
VM, ARM64 TARGET
Linux x86-64 7.23
macOS ARM64 21.6
VM, ARM32 TARGET
Linux x86-64 5.37
macOS ARM64 14.6
Audio decode: AAC in MP4
x realtime
CONTAINER
Linux x86-64 1,577
macOS ARM64 2,273
VM, HOST-NATIVE
Linux x86-64 1,285
macOS ARM64 2,162
VM, ARM64 TARGET
Linux x86-64 25.8
macOS ARM64 90.4
VM, ARM32 TARGET
Linux x86-64 33.4
macOS ARM64 79.1
Audio decode: AC-3 in MP2TS
x realtime
CONTAINER
Linux x86-64 1,663
macOS ARM64 2,636
VM, HOST-NATIVE
Linux x86-64 1,426
macOS ARM64 2,509
VM, ARM64 TARGET
Linux x86-64 30.6
macOS ARM64 104
VM, ARM32 TARGET
Linux x86-64 36.5
macOS ARM64 88.7
Audio decode: HE-AAC in MP2TS
x realtime
CONTAINER
Linux x86-64 1,042
macOS ARM64 1,363
VM, HOST-NATIVE
Linux x86-64 955
macOS ARM64 1,368
VM, ARM64 TARGET
Linux x86-64 14.5
macOS ARM64 39.0
VM, ARM32 TARGET
Linux x86-64 14.8
macOS ARM64 34.3

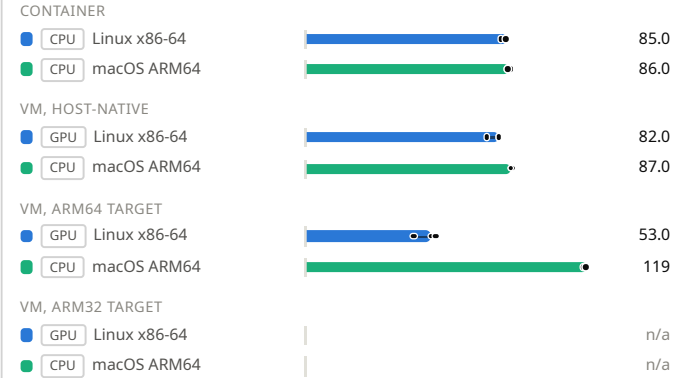
Audio decode: HE-AAC in MP4
x realtime



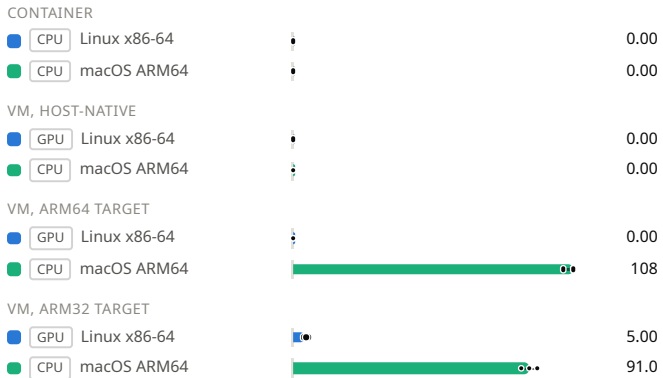
Audio decode: MP2 in MP2TS
x realtime



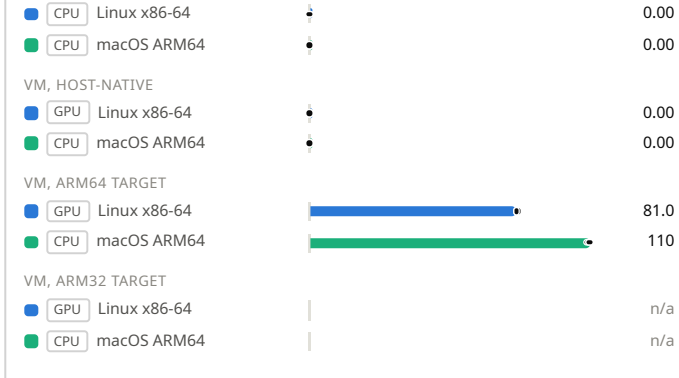
Frames dropped in playback: AVC in MP4, through Flutter · 1920x1080
frames · lower is better



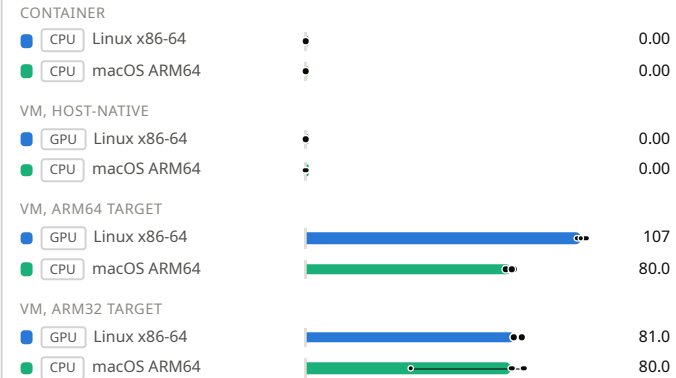
Frames dropped in playback: AVC in MP4, through OpenGL ES · 1920x1080
frames · lower is better



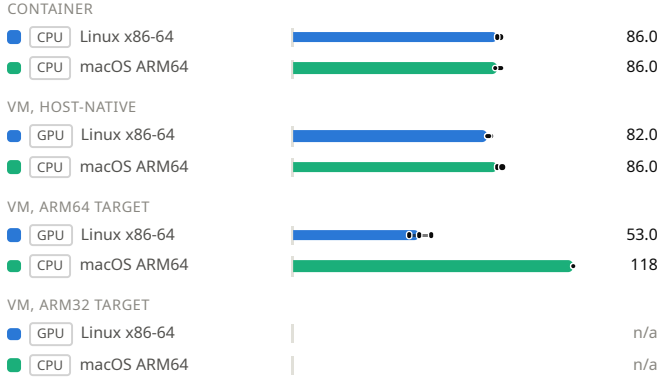
Frames dropped in playback: AVC in MP4, through Vulkan · 1920x1080
frames · lower is better



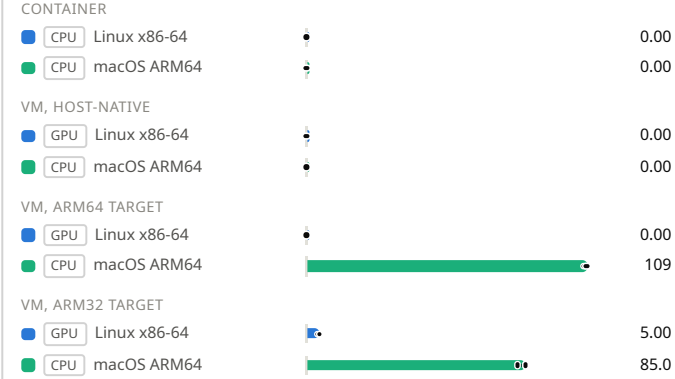
Frames dropped in playback: AVC in MP4, through Wayland · 1920x1080
frames · lower is better



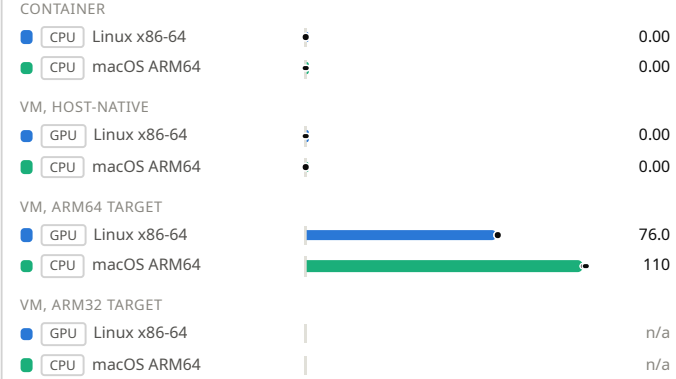
Frames dropped in playback: HEVC in MSE fragments, through Flutter · 1920x1080
frames · lower is better



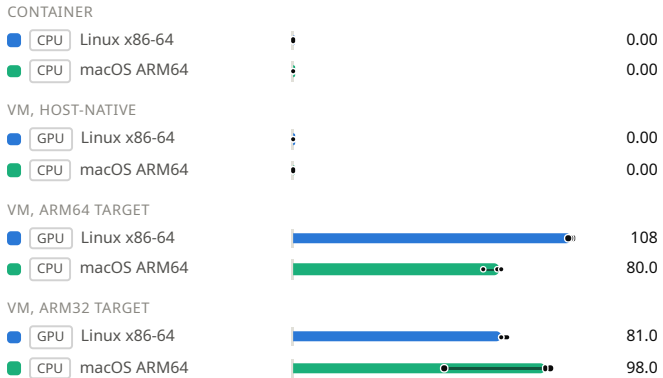
Frames dropped in playback: HEVC in MSE fragments, through OpenGL ES · 1920x1080
frames · lower is better



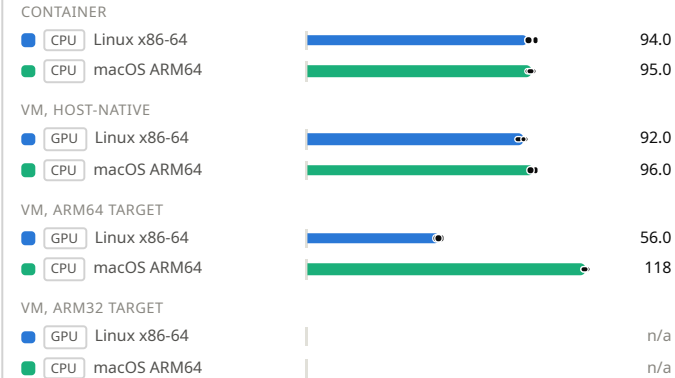
Frames dropped in playback: HEVC in MSE fragments, through Vulkan · 1920x1080
frames · lower is better



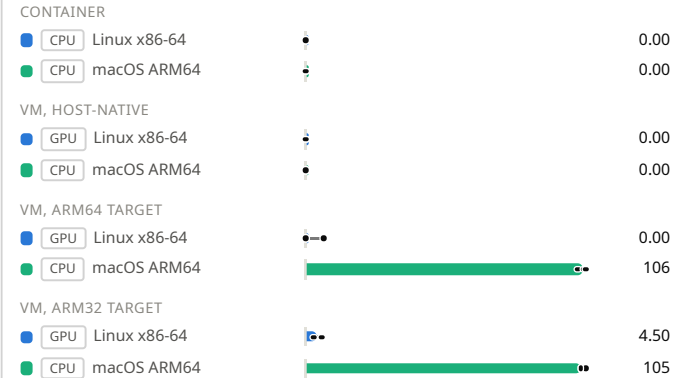
Frames dropped in playback: HEVC in MSE fragments, through Wayland · 1920x1080
frames · lower is better

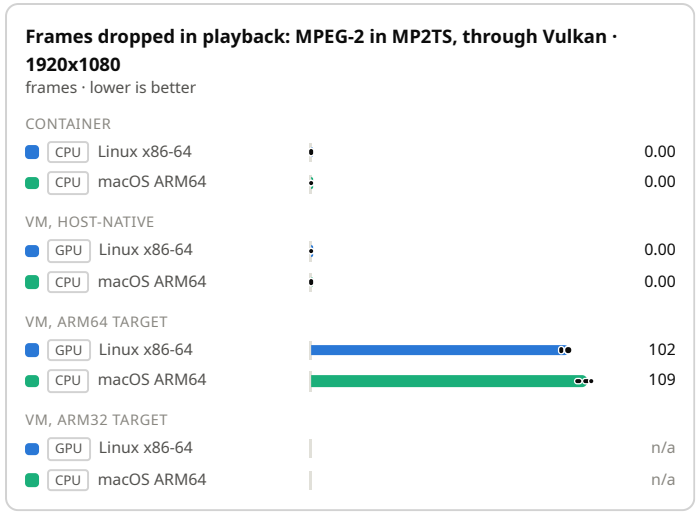


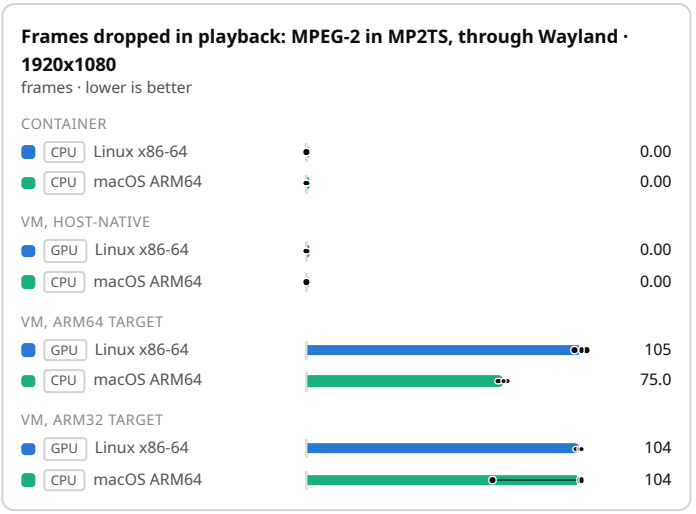
Frames dropped in playback: MPEG-2 in MP2TS, through Flutter · 1920x1080
frames · lower is better



Frames dropped in playback: MPEG-2 in MP2TS, through OpenGL ES · 1920x1080
frames · lower is better




Frames dropped in playback: MPEG-2 in MP2TS, through Vulkan · 1920x1080
frames · lower is better
CONTAINER
CPU Linux x86-64 0.00
CPU macOS ARM64 0.00
VM, HOST-NATIVE
GPU Linux x86-64 0.00
CPU macOS ARM64 0.00
VM, ARM64 TARGET
GPU Linux x86-64 102
CPU macOS ARM64 109
VM, ARM32 TARGET
GPU Linux x86-64 n/a
CPU macOS ARM64 n/a


Frames dropped in playback: MPEG-2 in MP2TS, through Wayland · 1920x1080
frames · lower is better
CONTAINER
CPU Linux x86-64 0.00
CPU macOS ARM64 0.00
VM, HOST-NATIVE
GPU Linux x86-64 0.00
CPU macOS ARM64 0.00
VM, ARM64 TARGET
GPU Linux x86-64 105
CPU macOS ARM64 75.0
VM, ARM32 TARGET
GPU Linux x86-64 104
CPU macOS ARM64 104